



# jQuery 101

**jQuery** is a fast, small, and feature-rich JavaScript library.

It's designed to ease HTML DOM manipulation, event handling, animation, and AJAX Http requests, with a simple API that works across the various browsers.

# jQuery Usage

Adding the jQuery library to an HTML page using a CDN:

```
<head>
  <script
    src="https://cdnjs.cloudflare.com/ajax/libs/jquery/2.2.4/jquery.min.js">
  </script>
</head>
```

The jQuery library exposes itself on the global scope

The library can be used through the **jQuery** property or  
with the commonly used **\$** alias

```
// define a callback function to run when the dom is ready
$(document).ready(function() {
  // ...
});
```

# DOM manipulation ...with jQuery

The jQuery API makes it really easy to do cross-browser DOM manipulation

```
// fetch element by id
var element = $('#id');

// fetch elements by class name
var elements = $('.className');

// fetch elements by tag type
var elements = $('p');

// fetch elements where the id attribute contains str
var elements = $('[id*=str]');

var element = '<tr>' + '<td>table entry</td>' + '</tr>';

// create the new element and append it to every node with the given class name
$(element).appendTo('.className');

// specify a click event callback function for every button
$('button').click(function(event){
    ...
});
```

{ Exercise }

## jQuery DOM manipulation

Substitute all DOM API manipulation on customer data exercise with jQuery

# AJAX ...with jQuery

Cross-browser AJAX requests are much simpler to achieve with the jQuery API

```
function successCallback(response) {  
    // do something with the data  
}  
  
function errorCallback(request, status, error) {  
    // do something with the error  
}  
  
// perform an ajax http get request  
$.ajax({  
    url: 'http://localhost:8080/api/users',  
    async: true,  
    success: successCallback,  
    error: errorCallback  
});
```

```
// perform an ajax http post request  
$.ajax({  
    url: 'http://localhost:8080/api/user',  
    type: 'POST',  
    data: JSON.stringify({  
        username: 'ferrao',  
        password: 'master',  
        email: 'rui.ferrao@academiadecodigo.org'  
    }),  
    async: true,  
    contentType: 'application/json',  
    success: successCallback,  
    error: errorCallback  
});
```

{ Exercise }

## AJAX with jQuery

Replace XMLHttpRequest with jQuery